

Exact Transfer of Bounded Linear Recovery and Relative Weight Hierarchies

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Abstract

Let $I \leq \mathbb{F}_q^E$, let $P \subseteq E$ be a target set, and put $J = E \setminus P$. The target and helper columns determine a nested pair $K_P \subseteq D_P \leq \mathbb{F}_q^J$. Its relative generalized Hamming weight $M_t(D_P, K_P)$ is the minimum helper union recovering t independent target combinations. We prove two exact transfer theorems. First, for an L -linear outer code $O \leq L^N$ with $N \geq 2$ and $d(O^\perp) > r + 1$, every such cost- $\leq r$ system is confined to its inner block exactly when

$$r < M_t(D_P, K_P) + d(I^\perp).$$

Below it, zero-extension preserves normalized equations and exact supports. Second, for one target coordinate in a block onto which O projects nontrivially, target-constrained inner coset weights and the functional dual of O give the exact finite nonconfinement cost without an outer-distance hypothesis. A single-parity-check family transfers beyond the ordinary outer support-distance gate.

Positive-density and bounded service-rate transfer follow. Equal relative-weight hierarchies need not determine bounded repair reliability, and presentations of the same nested pair can have different confinement thresholds. A projective-simplex family gives a non-MDS hierarchy and projective-rank reliability. The criterion $r + 1 < z_x(I)$ is the rank-one zero-functional case.

1 Introduction

Locality asks how many helpers recover one erased coordinate. Cooperative locality and (r, δ) -locality ask for recovery after several erasures, and availability asks for disjoint choices of helpers. Generalized Hamming weights give global lower bounds for cooperative recovery [17, 7, 15, 1]. Recovery sets and coordinate-indexed recovery structures record more local information [13], but support data alone discard the dual coefficients that implement the recovery equations.

This paper applies the relative generalized Hamming weights of the associated nested pair in the generator matrix to fixed-target helper costs. Relative generalized Hamming weights and the relative dimension/length profile were introduced for nested codes by Luo, Mitropant, Vinck, and Chen, who interpreted them as access thresholds in a wiretap model [11]; later work develops their ramp-secret-sharing and algorithmic roles [6, 18, 5]. The RGHW identity below is the standard nested-code support invariant applied to the pair induced by a represented target/helper split. The new content begins with concatenation: after the outer-dual-distance gate, each relative weight acquires the same additive inner-dual obstruction and every bounded normalized equation transfers, not only its numerical cost. At one target coordinate a finer weighted theorem removes the gate and gives the exact finite-length obstruction.

Let q be a prime power. Let $I \leq \mathbb{F}_q^E$ have a full-row-rank generator matrix $G = (G_P \mid G_J)$, where P is the target set and $J = E \setminus P$. Put

$$U_P = \text{im } G_P, \quad W_P = U_P \cap \text{im } G_J,$$

and define the associated nested code pair

$$K_P = \ker G_J \subseteq D_P = \{a \in \mathbb{F}_q^J : G_J a \in U_P\}.$$

We assume $0 < \dim I < |E|$, so both the represented message space and $I^\perp$ are nonzero. We use the convention $d(0) = \infty$ for the zero code. The quotient $D_P/K_P \cong W_P$ is exactly the space of target linear combinations recoverable from the helper coordinates. Here t is the dimension of a recovered subspace of W_P , not the number of erased coordinate positions. Recovering all values on P is the special case $U_P \subseteq \text{im } G_J$ and $t = \dim U_P$; dependent target columns may make this dimension smaller than $|P|$.

Theorem 1.1 (Main theorem: ranked transfer after the outer-distance gate). *Let $\ell = \dim W_P \geq 1$ and $1 \leq t \leq \ell$. The minimum cardinality of the union of helpers in a system recovering a t -dimensional subspace of W_P is*

$$\mu_t(P) = M_t(D_P, K_P).$$

Let I be used as the represented inner code in a concatenation with an L -linear outer code $O \leq L^N$, where L is the message space of I . For every nonnegative integer r , if $N \geq 2$ and $d(O^\perp) > r + 1$, then the concatenation confines every recovery system using at most r helpers for every internally recoverable t -dimensional target subspace to its target inner block if and only if

$$r < M_t(D_P, K_P) + d(I^\perp).$$

When the inequality holds, zero-extension gives a bijection on the normalized recovery-equation systems and their exact helper supports. Consequently, for any outer family with dual distance tending to infinity, the same statement holds for all sufficiently long concatenations at every fixed r .

The first assertion follows by identifying a recovery system with a subspace $A \leq D_P$ disjoint from K_P . Passing between this form and the standard definition of $M_t(D_P, K_P)$ uses a complement to $A \cap K_P$. The second assertion combines that identity with a fixed-demand dual decomposition: a nonconfined recovery system consists of its least inner realization together with a nonzero inner-dual perturbation in another block. The finite outer gate removes every nonzero outer-functional alternative; dual-distance growth makes that gate automatic at fixed radius.

Section 4 also retains the nonzero-functional sector in rank one. Target-constrained inner coset weights then give the exact finite minimum for every outer code whose projection onto the target block is nonzero, including cases where ordinary outer support distance does not decide transfer. These are two distinct levels of exactness: arbitrary recovered dimension after the outer-distance gate, and fully weighted finite-length exactness at one coordinate.

The same bijection yields the later transfer results. It preserves every block-local statistic determined by normalized recovery systems of cost at most r . After forgetting coefficients, it also preserves statistics determined by inclusion-minimal helper supports, including bounded repair reliability and the bounded service-rate region. This statement is block-local: the upward-closed recovery-set family in the full concatenated coordinate set also contains irrelevant cross-block supersets.

The theorem contains the previous one-coordinate criterion. If $P = \{x\}$ and $t = 1$, then $M_1(D_P, K_P)$ is the least number of helpers in a dual word through x , and

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp).$$

Thus $r < M_1 + d(I^\perp)$ is exactly $r + 1 < z_x(I)$.

The separation results delimit the information carried by this hierarchy:

relative-weight hierarchy $\not\Rightarrow$ bounded repair reliability, $(K_P, D_P) \not\Rightarrow$ confinement threshold.

The first failure comes from the overlap pattern of all recovery equations; the second comes from the coefficient presentation through $d(I^\perp)$.

Concatenation has long been used to construct LRCs with prescribed parameter properties [10, 8]. Jin and Fu fix a binary $[3, 2, 2]$ inner code and choose $\mathbb{F}_4$ outer codes to obtain parameter-optimal binary locality-two codes. Here the inner representation is arbitrary, and the question is the exact obstruction to preserving every bounded dual recovery equation: weighted and ungated in rank one, or rank-stratified after the outer-distance gate. We make no minimum-bandwidth claim.

2 Recovery sets and normalized equations

Let $C \leq \mathbb{F}_q^E$ be a linear code. We first separate four layers that are often harmlessly conflated when only locality is at issue.

Fix a target $x \in E$ and a helper bound r . A dual word $w \in C^\perp$ with $w_x \neq 0$ gives

$$c_x = \sum_{y \neq x} -\frac{w_y}{w_x} c_y \quad (c \in C).$$

Its exact helper support is $\text{supp}(w) \setminus \{x\}$. Define

$$\mathcal{H}_x^{(\leq r)}(C) = \{\text{supp}(w) \setminus \{x\} : w \in C^\perp, w_x \neq 0, \text{wt}(w) \leq r + 1\}.$$

The usual bounded recovery-set family is the upward closure of $\mathcal{H}_x^{(\leq r)}(C)$ inside the subsets of $E \setminus \{x\}$ of size at most r . Its inclusion-minimal members form the minimal recovery-set clutter. Thus exact supports, recovery sets, and minimal recovery sets are distinct but determine the same monotone success event.

The coefficient layer is

$$\mathcal{P}_x^{(\leq r)}(C) = \{w \in C^\perp : w_x = 1, \text{wt}(w) \leq r + 1\}.$$

These normalized recovery equations determine their exact supports, while the converse generally fails. The operational model is scalar exact repair: each participating helper supplies one symbol of $\mathbb{F}_q$. Array-code access, subpacketization, and repair bandwidth are different optimization problems [19, 21, 9].

If $A \subseteq E \setminus \{x\}$ is the surviving helper set, put

$$f_{x,r}(A) = 1 \iff A \text{ contains a member of } \mathcal{H}_x^{(\leq r)}(C).$$

Under independent helper survival with probabilities (s_y) , the bounded repair reliability is

$$R_{x,r}((s_y)) = \mathbb{E}[f_{x,r}].$$

It depends only on the minimal recovery sets, but it is not determined by their minimum cardinality. Section 6 shows that even the complete relative-weight hierarchy does not determine the corresponding bounded reliability law.

For several target coordinates, a recovery equation is a dual subspace rather than one dual word. Let $P \subseteq E$, $J = E \setminus P$, and let $G = (G_P \mid G_J)$ be a full-row-rank generator matrix. If

$T \leq \text{im } G_P$ is a subspace of target linear combinations, a normalized recovery system for T consists of linear maps

$$\alpha : T \longrightarrow \mathbb{F}_q^P, \quad \beta : T \longrightarrow \mathbb{F}_q^J$$

such that

$$G_P \alpha = \text{id}_T, \quad G_J \beta = -\text{id}_T.$$

For each $t \in T$, the vector $(\alpha(t), \beta(t))$ is then a dual word. The map α records the chosen target normalization and β records the helper coefficients.

The helper union of the system is the support of $\beta(T)$:

$$\text{supp}(\beta(T)) = \{j \in J : \text{some } t \in T \text{ has } \beta(t)_j \neq 0\}.$$

Its helper cost is $|\text{supp}(\beta(T))|$. We call a target subspace internally recoverable when it lies in

$$W_P = \text{im } G_P \cap \text{im } G_J.$$

Only this recovered subspace dimension is counted below. Relations supported entirely on P have recovered dimension zero and do not consume helpers.

For later comparison, the normalized equations at one coordinate reconstruct the represented code at radius r when their span is $C^\perp$. The least such r is the reconstruction radius.

Theorem 2.1 (MDS reconstruction from minimum-weight recovery equations). *Let $C \leq \mathbb{F}_q^E$ be an $[n, k]_q$ MDS code with $1 \leq k < n$, and let $x \in E$. Then*

$$\text{span } \mathcal{P}_x^{(\leq k)}(C) = C^\perp,$$

the reconstruction radius is k , and the supports of the minimum-weight recovery equations form the complete k -uniform clutter on $E \setminus \{x\}$.

Proof. The dual has dimension $n - k$ and minimum distance $k + 1$. For every $(k + 1)$ -set $T \subseteq E$, restriction of $C^\perp$ to $E \setminus T$ has domain dimension $n - k$ and codomain dimension $n - k - 1$. Its kernel contains a nonzero word supported on T , and the distance bound forces its support to be all of T . Hence the exact helper supports through x are precisely the k -subsets of $E \setminus \{x\}$.

Choose a fixed $(k - 1)$ -set $Q \subseteq E \setminus \{x\}$. For every $t \in E \setminus (\{x\} \cup Q)$, choose the normalized minimum dual word supported on $\{x, t\} \cup Q$. Evaluation at the private coordinates t shows that these $n - k$ words are independent. They are therefore a basis of $C^\perp$. No smaller radius contains a nonzero normalized equation, again by the dual-distance bound. $\square$

The support clutter in this theorem depends only on the MDS parameters. The normalized equations retain the represented dual code. This is the first instance of the support-coefficient distinction used in the transfer and separation results below.

3 Relative weights of the associated nested pair

Continue with $G = (G_P \mid G_J)$ and set

$$U_P = \text{im } G_P, \quad W_P = U_P \cap \text{im } G_J,$$

$$K_P = \ker G_J, \quad D_P = G_J^{-1}(U_P).$$

Proposition 3.1 (Associated nested code pair). *Restriction of G_J gives an exact sequence*

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0. \quad (1)$$

Proof. By definition, $K_P = \ker G_J$ is contained in $D_P = G_J^{-1}(U_P)$. The kernel of the restricted map is therefore K_P . Its image is

$$G_J(D_P) = \text{im } G_J \cap U_P = W_P,$$

so the restricted map is onto W_P . $\square$

The first isomorphism theorem applied to the restricted map gives $D_P/K_P \cong W_P$.

The pair depends on the represented generator matrix and the target set. A change of message basis gives an isomorphic pair; monomial relabeling of helpers gives the corresponding monomially equivalent pair.

For a subspace $A \leq \mathbb{F}_q^J$, write $\text{supp } A$ for the union of the supports of its vectors. For nested linear codes $K \subseteq D \leq \mathbb{F}_q^J$, the t th relative generalized Hamming weight is

$$M_t(D, K) = \min\{|\text{supp } L| : L \leq D, \dim L - \dim(L \cap K) = t\}.$$

Equivalently, one may minimize over subspaces disjoint from K and of dimension t . Strict growth and the relative Singleton bound are Proposition 2 and Theorem 4 of Luo et al. [11]; standard treatments also cover ordinary Wei duality and computational forms [6, 18].

Theorem 3.2 (Relative weights are exact recovery costs). *Let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The minimum helper-union size among recovery systems for all internally recoverable t -dimensional subspaces of target linear combinations is*

$$\mu_t(P) = M_t(D_P, K_P).$$

More precisely, normalized recovery systems of recovered dimension t are in support-preserving correspondence, after forgetting the target normalization, with subspaces

$$L \leq D_P, \quad \dim L = t, \quad L \cap K_P = 0,$$

while the target normalization is a linear section through G_P .

Proof. Let a normalized system recover a t -dimensional subspace $T \leq W_P$. In the notation of Section 2, put $L = \beta(T)$. Since $G_J\beta = -\text{id}_T$, the map β is injective, $L \leq D_P$, and $L \cap K_P = 0$. Moreover $G_JL = T$, and the union of helpers in the system is exactly $\text{supp } L$.

Conversely, let $L \leq D_P$ have dimension t and meet K_P trivially. Then $G_J|_L$ is an isomorphism from L onto the t -dimensional subspace $T = G_JL \leq W_P$. Choose a linear section $\alpha : T \rightarrow \mathbb{F}_q^P$ with $G_P\alpha = \text{id}_T$. Put $\beta = -(G_J|_L)^{-1}$. Then $G_P\alpha + G_J\beta = 0$, so (α, β) is a normalized recovery system with helper union $\text{supp } L$.

It remains to compare this minimum with the standard relative-weight definition. If $L' \leq D_P$ satisfies

$$\dim L' - \dim(L' \cap K_P) = t,$$

choose a vector-space complement L to $L' \cap K_P$ inside L' . Then $\dim L = t$, $L \cap K_P = 0$, and $\text{supp } L \subseteq \text{supp } L'$. The reverse inclusion of feasible classes is immediate. The two minima are equal. $\square$

The corresponding relative dimension/length profile is

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H|=s}} (\dim(D_P \cap \mathbb{F}_q^H) - \dim(K_P \cap \mathbb{F}_q^H)),$$

where $\mathbb{F}_q^H$ denotes vectors supported on H .

Proposition 3.3 (Recovery interpretation of the relative profile). *For $0 \leq s \leq |J|$,*

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H|=s}} \dim(W_P \cap \text{span}(G_H)).$$

Consequently,

$$M_t(D_P, K_P) = \min\{s : K_s(D_P, K_P) \geq t\}.$$

Thus K_s is the maximum number of independent target linear combinations recoverable from a fixed budget of s helpers.

Proof. Fix H . Restrict the exact map (1) to $D_P \cap \mathbb{F}_q^H$. Its kernel is $K_P \cap \mathbb{F}_q^H$, and its image is $W_P \cap \text{span}(G_H)$. Rank-nullity gives the first equality after maximizing over H . The second is the standard inverse relation between the relative dimension/length profile and relative generalized weights; it also follows directly from Theorem 3.2. $\square$

The shortening–puncturing duality behind the relative profiles of nested codes is standard in wiretap and ramp-secret-sharing theory [11, Lemma 1 and Theorem 2]; see also [6, eqs. (1)–(2)]. Its recovery interpretation gives the failure-side counterpart of Theorem 3.2.

Proposition 3.4 (Failures and residual ambiguity). *Let $\ell = \dim W_P$. If the helpers in $H \subseteq J$ survive and those in $F = J \setminus H$ fail, then the unrecovered target dimension is*

$$\ell - \dim(W_P \cap \text{span}(G_H)) = \dim(K_P^\perp \cap \mathbb{F}_q^F) - \dim(D_P^\perp \cap \mathbb{F}_q^F). \quad (2)$$

Consequently, for $1 \leq t \leq \ell$,

$$M_t(K_P^\perp, D_P^\perp) = \min\{|F| : \dim(W_P \cap \text{span}(G_{J \setminus F})) \leq \ell - t\}. \quad (3)$$

Thus the dual relative weight is the minimum number of helper failures that leave at least t target dimensions ambiguous.

Proof. For a code $C \leq \mathbb{F}_q^J$, puncturing onto F gives

$$\dim C - \dim(C \cap \mathbb{F}_q^H) = \dim(C|_F) = |F| - \dim(C^\perp \cap \mathbb{F}_q^F).$$

Apply this identity to D_P and K_P and subtract. Since $\dim D_P - \dim K_P = \ell$, Proposition 3.3 gives (2). The right-hand side is the relative dimension of the dual pair $D_P^\perp \subseteq K_P^\perp$ on F . Its least support size at value at least t is $M_t(K_P^\perp, D_P^\perp)$: a subspace attaining the relative weight is supported on such an F , while any relative dimension at least t admits a t -dimensional subspace disjoint from $D_P^\perp \cap \mathbb{F}_q^F$. This proves (3). $\square$

The strict inequalities

$$1 \leq M_1(D_P, K_P) < \cdots < M_\ell(D_P, K_P)$$

therefore say that each additional independent target combination requires a strictly larger helper union. The next section shows that concatenation adds the same inner-dual cost $d(I^\perp)$ to every level of this hierarchy.

4 Exact confinement under concatenation

Let $L/\mathbb{F}_q$ be a finite extension and let $\iota : L \xrightarrow{\sim} I \leq \mathbb{F}_q^E$ be an $\mathbb{F}_q$ -linear isomorphism, used as the inner encoder. Assume $0 < \dim I < |E|$. For an L -linear outer code $O \leq L^N$, write

$$O \circ I = \{(\iota(a_1), \dots, \iota(a_N)) : (a_1, \dots, a_N) \in O\}.$$

The proof of confinement uses the linear map

$$\Phi_I : \mathbb{F}_q^E \longrightarrow L^*, \quad \Phi_I(y)(a) = \langle y, \iota(a) \rangle.$$

Here $L^* = \text{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$. Its kernel is $I^\perp$. A block vector $y = (y_1, \dots, y_N)$ belongs to $(O \circ I)^\perp$ precisely when

$$(\Phi_I(y_1), \dots, \Phi_I(y_N)) \in \text{FD}(O), \quad (4)$$

where

$$\text{FD}(O) = \{(\beta_j) \in (L^*)^N : \sum_j \beta_j(a_j) = 0 \text{ for every } (a_j) \in O\}.$$

Indeed, both conditions say that y is orthogonal to every encoded outer word. Under the trace identification $L \cong L^*$, L -linearity of O identifies $\text{FD}(O)$ with $O^\perp$ and preserves block support. Thus its support distance is $d(O^\perp)$.

Fix a target block and a nonzero internally recoverable subspace $T \leq W_P$. Let $\rho_T(I)$ be the minimum helper-union size of a normalized recovery system for T in the inner code. In a concatenation, a normalized system for T means a linear family of dual equations whose target-block coefficient map $\alpha : T \rightarrow \mathbb{F}_q^P$ satisfies $G_P \alpha = \text{id}_T$ under the fixed inner-message identification. This agrees with the intrinsic target image of the concatenated generator whenever the outer projection onto the chosen block is surjective. In particular, $d(O^\perp) > 1$ guarantees surjectivity: an L -linear coordinate image is either 0 or L , and a zero image would put a weight-one word in $O^\perp$. A system is *confined* if every one of its equations is supported in the target block.

Theorem 4.1 (Confinement for a fixed target subspace). *Let r be a nonnegative integer and let $O \leq L^N$ be an L -linear outer code with $N \geq 2$ and $d(O^\perp) > r + 1$. Then $O \circ I$ confines every normalized recovery system for T of cost at most r if and only if*

$$r < \rho_T(I) + d(I^\perp). \quad (5)$$

Below this bound, zero-extension gives a support-preserving bijection between the inner systems of cost at most r and the corresponding systems in the concatenation. Hence the same conclusions hold for all sufficiently long members of any outer family with dual distance tending to infinity, at every fixed r .

Proof. Apply (4) to every equation in a recovery system. If one of the resulting functional tuples is nonzero, its block support has size at least $d(O^\perp)$. A system with helper bound r can meet only a bounded number of blocks: every nonzero block outside the target block contains a helper, while the target block contributes at most one further block. A cost- $\leq r$ system therefore meets at most $r + 1$ blocks. Hence $d(O^\perp) > r + 1$ excludes this case.

Every remaining block component lies in $I^\perp$. The target-block component is an inner recovery system for T , so its helper union has size at least $\rho_T(I)$. If the system is not confined, some equation map in another block is nonzero. Its image is a nonzero subspace of $I^\perp$ and therefore has union support at least $d(I^\perp)$. Disjoint blocks make the two costs additive. Every nonconfined system consequently costs at least $\rho_T(I) + d(I^\perp)$.

This lower bound is attained. Take a minimum inner system for T , a minimum-weight word $v \in I^\perp$, and a nonzero linear functional $\lambda : T \rightarrow \mathbb{F}_q$. In a second block add the rank-one equation map $a \mapsto \lambda(a)v$. Both block-functional images vanish, so the new system belongs to the concatenated dual; it is nonconfined and has the stated cost. This proves both directions of (5). Below the bound every system is confined, and restriction to the target block is inverse to zero-extension. $\square$

Minimizing the fixed-subspace threshold over all recovered subspaces of fixed dimension gives the promised hierarchy.

Theorem 4.2 (Confinement by recovered dimension after the outer-distance gate). *Let r be a nonnegative integer, let $O \leq L^N$ be L -linear with $N \geq 2$ and $d(O^\perp) > r + 1$, and let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The concatenation $O \circ I$ confines every cost- $\leq r$ recovery system for every internally recoverable t -dimensional target subspace if and only if*

$$r < M_t(D_P, K_P) + d(I^\perp). \quad (6)$$

When (6) holds, restriction and zero-extension preserve every normalized equation and its exact helper support. For an outer family with dual distance tending to infinity, these conclusions hold for all sufficiently long members at every fixed r .

Proof. By Theorem 3.2, every such target subspace has inner cost at least $M_t(D_P, K_P)$, and some target subspace attains this minimum. Apply Theorem 4.1 to each subspace. The lower bound gives sufficiency uniformly; an attaining subspace and the external rank-one perturbation in that proof give necessity. $\square$

4.1 An ungated finite formula for one target coordinate

The block-functional decomposition also gives a fully weighted finite-length criterion in rank one. For $\beta \in L^*$, put

$$\lambda_I(\beta) = \min_{\Phi_I(y)=\beta} \text{wt}(y), \quad \mu_{I,x}(\beta) = \min_{\substack{\Phi_I(y)=\beta \\ y_x \neq 0}} \text{wt}(y),$$

with value ∞ when a constrained fiber is empty. The first quantity is the usual coset-leader weight for the inner encoder. The second retains the target-coordinate constraint. The functional-dual decomposition itself is classical concatenation machinery [3, Theorem 2.1].

Theorem 4.3 (Exact weighted finite-length confinement at one coordinate). *Fix an outer block j , an inner coordinate x , and $N \geq 2$, and assume that the coordinate projection $O \rightarrow L$ at block j is nonzero, hence surjective. The minimum total Hamming weight of a word in $(O \circ I)^\perp$ that is nonzero at (j, x) and is not the zero-extension of a word of $I^\perp$ in block j is*

$$Z_{j,x}(O, I) = \min \left\{ \mu_{I,x}(0) + d(I^\perp), \min_{\substack{\beta=(\beta_h) \in \text{FD}(O) \\ \beta \neq 0}} \left(\mu_{I,x}(\beta_j) + \sum_{h \neq j} \lambda_I(\beta_h) \right) \right\}. \quad (7)$$

A minimum over an empty set is ∞ . Consequently, for every $r \geq 0$, restriction to block j and zero-extension are inverse, support-preserving bijections between the normalized recovery equations through x in I and those through (j, x) in $O \circ I$, both using at most r helpers, if and only if

$$Z_{j,x}(O, I) > r + 1. \quad (8)$$

The same equivalence holds for the exact helper-support families. When these conditions hold, the inclusion-minimal recovery sets are copied as well.

Proof. Write a concatenated dual word in blocks as $y = (y_h)_{h=1}^N$ and put $\beta_h = \Phi_I(y_h)$. By (4), the blocks lie in independent fibers indexed by one tuple $\beta \in \text{FD}(O)$. For fixed β , the least target-block weight subject to $y_j(x) \neq 0$ is $\mu_{I,x}(\beta_j)$, and the least weight in every other block is $\lambda_I(\beta_h)$. Thus a nonzero functional tuple has the second cost in (7). Such a word cannot be an embedded inner-dual word, because the latter induces the zero functional in every block.

Surjectivity of $O \rightarrow L$ at block j also shows that a dual word confined to block j has $\beta_j = 0$. Thus the embedded inner-dual words are exactly the confined words. If $\beta = 0$, every block lies in $I^\perp$. A word through (j, x) that is not confined to block j pays at least $\mu_{I,x}(0)$ in the target block and $d(I^\perp)$ in another block. Placing minimizers in two blocks attains their sum. The zero and nonzero functional sectors exhaust all nonconfined words, proving (7).

A normalized recovery equation with at most r helpers has total weight at most $r+1$. Hence (8) excludes every nonconfined equation at that radius. Restriction and zero-extension then give the asserted bijections. Conversely, if $Z_{j,x}(O, I) \leq r+1$, normalize an attaining word at (j, x) ; it is a nonconfined recovery equation using at most r helpers. This proves the equivalence. $\square$

Theorem 4.3 retains information that the outer support distance discards. The following family makes the difference strict without using a geometric inner code.

Proposition 4.4 (A family from a Singer cycle beyond the outer-distance gate). *Let $k \geq 2$ and suppose*

$$\frac{q^k - 1}{q - 1} > (k + 1)^2. \quad (9)$$

Let I be the $[k + 1, k, 2]_q$ single-parity-check code generated by

$$G = (I_k \mid \mathbf{1}),$$

and identify its message space with $L = \mathbb{F}_{q^k}$. For every $k + 1 \leq n \leq 2k + 1$, there is an L -linear length- n outer code O with $d(O^\perp) = n$ such that (8) holds at helper radius $n - 1$ for every inner coordinate and every outer block. Thus all normalized radius- $(n - 1)$ recovery equations and their exact helper supports are copied blockwise although the ordinary outer-distance gate fails.

In particular, this holds for the $[4, 3, 2]_5$ inner code and a length-five outer code at helper radius four.

Proof. The $k + 1$ coordinate functionals of the encoder give a set S of $k + 1$ points in the projective space on L^* . The multiplication maps of L , modulo $\mathbb{F}_q^*$, induce on that projective space a Singer cycle of order $(q^k - 1)/(q - 1)$, acting regularly on its points [20]. At most $|S|^2 = (k + 1)^2$ elements of the cycle send a point of S to a point of S , so (9) permits a multiplication map $a : L \rightarrow L$, explicitly $a(u) = \gamma u$ for some $\gamma \in L^\times$, whose dual action satisfies $a^*(S) \cap S = \emptyset$.

Set

$$O = \{u \in L^n : u_0 + a(u_1) + u_2 + \cdots + u_{n-1} = 0\}.$$

Every functional-dual tuple has the form $(f, f \circ a, f, \dots, f)$. If $f \neq 0$, all n entries are nonzero, so $d(O^\perp) = n$. A realization of total weight n would have weight one in every block. Weight-one realizations are exactly the scalar multiples of the $k + 1$ coordinate functionals, so both $[f]$ and $[f \circ a]$ would lie in S , contrary to the choice of a . Every nonzero tuple therefore costs at least $n + 1$.

Every outer coordinate projection is surjective. Finally, $I^\perp$ is spanned by $(1, \dots, 1, -1)$, so $d(I^\perp) = \mu_{I,x}(0) = k + 1$ for every x . The zero-functional cost in (7) is $2k + 2 > n$, and the nonzero-functional cost is at least $n + 1$. Thus $Z_{j,x}(O, I) > n$ for every j, x . At $r = n - 1$ the ordinary gate would require $d(O^\perp) > n$, which is false, while the exact weighted gate holds. Taking $(q, k, n) = (5, 3, 5)$ gives the stated concrete instance. Since the inner recovery equations have k helpers and $n - 1 \geq k$, the asserted transfer is nonvacuous throughout the family. $\square$

For a nondegenerate singleton target $P = \{x\}$, the first relative weight is the least number of helpers in a dual word through x . If the total-weight threshold used for one-coordinate equations is denoted by $z_x(I)$, then

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp). \quad (10)$$

The helper-radius condition (6) is therefore $r + 1 < z_x(I)$, with no change of convention hidden in the reduction. Under the outer-distance gate, every nonzero tuple in $\text{FD}(O)$ already meets more than $r + 1$ blocks, so Theorem 4.3 reduces to this zero-functional condition. Thus the ranked theorem is the clean higher-dimensional result after the nonzero-functional sector is excluded, while Theorem 4.3 gives the ungated finite criterion in rank one.

5 Consequences of the threshold

The fixed-target relative weights sit between ordinary generalized weights and worst-case cooperative locality. Let $d_e(C^\perp)$ denote the e th generalized Hamming weight of $C^\perp$. For an e -set $P \subseteq E$, let $\kappa_C(P)$ be the least $|H|$ for which some e -dimensional subcode $A \leq C^\perp$, supported on $P \cup H$, restricts isomorphically to $\mathbb{F}_q^P$. Set $\kappa_C(P) = \infty$ when no such subcode exists. Thus $\kappa_C(P)$ is the minimum number of helpers needed to recover all coordinates in P simultaneously.

Theorem 5.1 (Best-target generalized weight and first escape cost). *For $1 \leq e \leq \dim C^\perp$,*

$$\min_{\substack{P \subseteq E \\ |P|=e}} \kappa_C(P) = d_e(C^\perp) - e. \quad (11)$$

Consequently, if every e -set is recoverable and $r_e^{\text{coop}}(C) = \max_{|P|=e} \kappa_C(P)$, then

$$r_e^{\text{coop}}(C) \geq d_e(C^\perp) - e.$$

For a fixed inner code in the eventual concatenation regime of Theorem 4.1, the least first nonconfinement cost over all e -coordinate target sets is

$$d_e(C^\perp) - e + d(C^\perp). \quad (12)$$

Proof. A normalized identity system on P spans an e -dimensional subcode of $C^\perp$ whose support is P together with its helper union. Hence $d_e(C^\perp) \leq e + \kappa_C(P)$.

Conversely, let $A \leq C^\perp$ have dimension e and support size $d_e(C^\perp)$. An information set P of a generator matrix of A has size e , and restriction $A \rightarrow \mathbb{F}_q^P$ is an isomorphism. Row reduction on these columns gives normalized identity equations whose helper union is $\text{supp}(A) \setminus P$. This proves (11); the cooperative-locality inequality follows by taking the maximum instead of the minimum.

It remains to justify the escape formula when the target columns indexed by P are dependent. Put $U = \text{im } G_P$. A normalized identity system on $\mathbb{F}_q^P$ and a normalized recovery system on U have the same minimum helper union. Indeed, restricting an identity system along a linear section $\alpha : U \rightarrow \mathbb{F}_q^P$ gives a system on U . Conversely, given a system on U , write

$$a = \alpha(G_P a) + (a - \alpha(G_P a)) \quad (a \in \mathbb{F}_q^P).$$

The second summand lies in $\ker G_P$ and hence is a target-only dual equation, so adjoining these kernel equations recovers the full identity without adding helpers. Thus the common minimum is $\kappa_C(P)$.

Now repeat the block-functional argument of Theorem 4.1. Once the outer-functional case is excluded, the target block of any identity system uses at least $\kappa_C(P)$ helpers. If the system is nonconfined, a nonzero equation map in another block has union support at least $d(C^\perp)$; the two supports are disjoint. Conversely, attach a minimum-weight word of $C^\perp$ in a second block through any nonzero linear functional on $\mathbb{F}_q^P$. Hence the first nonconfinement cost for this P is $\kappa_C(P) + d(C^\perp)$. Minimizing over P and using (11) proves (12). $\square$

Put $k = \dim I$ and $b = \text{rank } G_J$. The exact sequence and the relative and ordinary Singleton bounds give

$$M_t(D_P, K_P) \leq b - \ell + t, \quad d(I^\perp) \leq k + 1. \quad (13)$$

Theorem 5.2 (MDS thresholds and rigidity). *For $1 \leq t \leq \ell$,*

$$M_t(D_P, K_P) + d(I^\perp) \leq 2k - \ell + t + 1. \quad (14)$$

Suppose I is MDS, and put

$$u = \min\{k, |P|\}, \quad b = \min\{k, |J|\}.$$

Then $\ell = u + b - k$. If $\ell \geq 1$, then, for $1 \leq t \leq \ell$,

$$M_t(D_P, K_P) = k - u + t, \quad M_t(D_P, K_P) + d(I^\perp) = 2k - u + t + 1. \quad (15)$$

If $|J| \geq k$, then $\ell = u$ and these values attain the ceiling in (14) at every rank. Conversely, when $\ell \geq 1$, equality in (14) at $t = 1$ forces I to be MDS and forces equality at every rank.

Proof. The ceiling follows from (13) and $b \leq k$. For an MDS code the column matroid is uniform, so $\text{rank } G_P = u$, $\text{rank } G_J = b$, $\ell = u + b - k$, and, for $H \subseteq J$ with $|H| = h \leq k$,

$$\dim(\text{span } G_P \cap \text{span } G_H) = \max\{0, u + h - k\}.$$

Thus Proposition 3.3 gives $M_t = k - u + t$ for $1 \leq t \leq \ell$; here $k - u + t \leq b \leq |J|$. Now $d(I^\perp) = k + 1$, and if $|J| \geq k$, then $b = k$ and $\ell = u$, proving the remaining direct assertions.

Equality at $t = 1$ forces equality in both Singleton bounds and $b \leq k$. Hence $d(I^\perp) = k + 1$, so I is MDS, and $M_1 = b - \ell + 1$. Strict growth then gives $M_t \geq b - \ell + t$, while relative Singleton gives the reverse inequality; every rank is extremal. $\square$

We next record the global parameters and the density of the transferred data. Let the inner code have parameters $[m, k, d]_q$, so $[L : \mathbb{F}_q] = k$. Let $K_N = \dim_L O_N$ and $D_N = d(O_N)$.

Corollary 5.3 (Positive-density realization). *Assume $d(O_N^\perp) \rightarrow \infty$. Below any fixed threshold supplied by Theorem 4.1 or Theorem 4.2, for all sufficiently large N every inner normalized recovery system and exact helper support is copied in each of the N blocks. Each fixed inner coordinate type has density $1/m$ among the concatenated coordinates; the union of the types in P has density p/m .*

The concatenated code has length mN , dimension kK_N , and minimum distance at least dD_N . If

$$\frac{K_N}{N} \longrightarrow R > 0, \quad \liminf \frac{D_N}{N} \geq \delta > 0,$$

then the concatenated family has rate kR/m and relative distance at least $d\delta/m$. Outer families satisfying all three hypotheses exist over every finite field L .

Proof. The transfer statement is the zero-extension bijection in the confinement theorems, applied independently in every block. There are mN coordinates, and each inner coordinate occurs once in each block, which gives the density claims. The blockwise inner encoder is injective, so restriction of scalars gives dimension kK_N . A nonzero outer word has at least D_N nonzero symbols, each encoding to an inner word of weight at least d . This proves the distance and asymptotic assertions.

For the existence statement, write $Q = |L|$. Choose $0 < R < 1$ and positive $\delta, \delta^\perp$ such that

$$H_Q(\delta) < 1 - R, \quad H_Q(\delta^\perp) < R,$$

where H_Q is the Q -ary entropy function. The first-moment argument for a random L -linear code shows that the expected numbers of nonzero primal and dual words below the two relative-weight cutoffs tend to zero. Hence one may choose a sequence with rate tending to R , primal relative distance at least δ , and dual relative distance at least $\delta^\perp$ [12, Chapter 17]. $\square$

Exact support transfer also preserves the fractional load-balancing region used in service-rate problems. We state the consequence because it requires only the inclusion-minimal recovery sets, whereas the theorem transfers all normalized coefficients as well.

Fix one inner target block with helper set J . Let $\mathcal{A}$ be a finite set of demands on that block, and let $\mathcal{M}_a^{(\leq r)}$ be the inclusion-minimal exact helper supports of size at most r for $a \in \mathcal{A}$. For a nonnegative helper-capacity vector c , define

$$\Lambda_{\mathcal{A}}^{(\leq r)}(c) = \left\{ \lambda \geq 0 : \begin{array}{l} \text{there are } f_{a,R} \geq 0 \text{ with } \lambda_a = \sum_{R \in \mathcal{M}_a^{(\leq r)}} f_{a,R}, \\ \sum_a \sum_{R \ni j} f_{a,R} \leq c_j \text{ for every helper } j \end{array} \right\}.$$

This is the bounded service-rate region of the recovery-set system [14, 4].

Corollary 5.4 (Transfer of bounded service-rate regions). *Let $O \leq L^N$ be an outer code with $d(O^\perp) > r + 1$. Suppose the radius r lies below the inner confinement threshold for every demand in $\mathcal{A}$. Transport capacities from the inner helper set to its copy in a target block of $O \circ I$. Then the inner and concatenated bounded service-rate regions are equal. Capacities assigned outside the target block do not change this equality. Consequently, for an outer family with $d(O_N^\perp) \rightarrow \infty$, the equality holds for all sufficiently large N .*

Proof. $d(O^\perp) > r + 1$ excludes every nonzero outer-functional tuple in a cost- $\leq r$ system, and the inner threshold excludes the remaining zero-functional nonconfined systems. Thus restriction and zero-extension copy every inclusion-minimal support and create no cross-block minimal support. The helper bijection therefore transports the feasible variables $f_{a,R}$ and every capacity inequality in both directions. Allowing the full upward-closed recovery-set family gives the same region: flow assigned to a nonminimal set can be moved to a contained minimal set and weakly decreases every helper load. Thus cross-block supersets and capacities outside the target block are irrelevant. $\square$

6 What the relative-weight hierarchy forgets

Relative generalized weights record minimum support sizes. They do not record the overlap pattern of all recovery sets, which controls stochastic repair. Nor do they record the target coefficient presentation, which can change $d(I^\perp)$ and hence the sum $M_t(D_P, K_P) + d(I^\perp)$. We give one separation of each kind.

First consider a one-dimensional recovered space. On the helper set $\{0, \dots, 8\}$, let

$$\begin{aligned}\mathcal{A} &= \{158, 026, 045, 013, 478\}, \\ \mathcal{B} &= \{168, 237, 078, 015, 124\},\end{aligned}$$

where, for example, 158 denotes $\{1, 5, 8\}$. Both are realizable as the minimum recovery sets at a distinguished coordinate of a represented $[10, 4, 6]_q$ code over a common finite prime field.

Proposition 6.1 (Rank-one reliability separation). *The two distinguished coordinates above have the same minimum helper cost $M_1 = 3$. Under independent homogeneous helper survival with probability s , their radius-three repair reliabilities are*

$$R_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \quad (16)$$

$$R_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8. \quad (17)$$

In particular, the complete rank-one relative-weight hierarchy does not determine bounded repair reliability.

Proof. The five triples in each clutter are the minimum recovery sets, so both minimum costs are three. For $k = 1, \dots, 5$, the numbers of k -edge subfamilies according to union size are

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9

Inclusion–exclusion for the five edge-containment events gives (16) and (17).

We now give the representation argument. Over an infinite field K , choose five projective lines $\ell_1, \dots, \ell_5$. For $\mathcal{A}$, make ℓ_2, ℓ_3, ℓ_4 concurrent and label

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5.$$

Choose 2, 6 on ℓ_2 , 3 on ℓ_4 , and 7 on ℓ_5 . For $\mathcal{B}$, make ℓ_1, ℓ_4, ℓ_5 concurrent at 1, put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose 6, 3, 5, 4 on $\ell_1, \ell_2, \ell_4, \ell_5$, respectively. At each free choice, avoid the finitely many unwanted joining lines. The displayed triples are then exactly the collinear triples.

Choose representatives $p_i \in K^3$ of either plane configuration, let $x = (1, 0, 0, 0)$, and lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the lifts over any displayed line and dependence of any four helper lifts are proper linear conditions on the t_i . For a displayed line, its three quotient vectors have a unique dependence up to scalar, and the same coefficients give one nonzero linear equation in the t_i . For four helper points, the quotient vectors span K^3 , so their lifted determinant is a nonzero linear form in the four first coordinates. Avoiding this finite union makes every helper triple and quadruple independent, while the dependent four-sets through x are exactly $\{x\} \cup A$ for the five prescribed triples A . Choose the construction over $\mathbb{Q}$ and reduce both configurations modulo one prime avoiding the finitely many nonzero determinants. The two row-span codes then have rank four, dual distance four, and minimum recovery sets exactly $\mathcal{A}$ and $\mathcal{B}$. A hyperplane not containing x contains at most three helpers because every four helpers are independent. A hyperplane containing x contains at most one displayed collinear triple, and such a triple gives four columns in a hyperplane. Thus the maximum hyperplane intersection has size four, so both codes have distance $10 - 4 = 6$. $\square$

The rank-one separation extends to every quotient rank without changing any relative generalized weight. For a nested pair $K \subseteq D \leq \mathbb{F}_q^J$ with $\dim(D/K) = \ell$, define its radius- R full-quotient reliability by

$$\Pr\{\exists L \leq D : L + K = D, |\text{supp } L| \leq R, \text{supp } L \subseteq S\},$$

where $S \subseteq J$ is the random set of independently surviving helpers, each present with probability s .

Theorem 6.2 (Equal relative weights, different reliability for full recovery). *For every $\ell \geq 1$, there are two nested code pairs of quotient dimension ℓ with identical relative generalized Hamming weights $M_1, \dots, M_\ell$ but different bounded reliability laws for recovery of the full quotient.*

Proof. For $\ell = 1$, use Proposition 6.1. For $\ell > 1$, let $K_0 \subset D_0$ be either of its two associated rank-one pairs. On disjoint helper blocks, add the same $\ell - 1$ pairs

$$0 \subset \langle \mathbf{1}_{L_i} \rangle \quad (1 \leq i \leq \ell - 1),$$

where each padding line has the unique support of size L_i . Thus

$$K = K_0 \oplus 0 \oplus \dots \oplus 0, \quad D = D_0 \oplus \langle \mathbf{1}_{L_1} \rangle \oplus \dots \oplus \langle \mathbf{1}_{L_{\ell-1}} \rangle.$$

For a direct sum of rank-one quotient pairs, the t th relative weight is the sum of the t smallest component costs. Both constructions therefore have the same hierarchy, obtained from the common multiset $\{3, L_1, \dots, L_{\ell-1}\}$.

To verify the direct-sum formula, project a feasible helper subspace to each quotient component. A t -dimensional quotient image has nonzero projection to at least t of the one-dimensional components. On disjoint helper blocks, each active projection pays at least that component's minimum cost, so the total support is at least the sum of the t smallest costs. The direct sum of minimum words from the t cheapest components attains the bound.

Set $R = 3 + \sum_i L_i$. A recovery subspace of full quotient rank projects nontrivially to every padding component and hence uses its entire forced support. Removing those supports leaves a radius-three recovery set in the base component. Conversely, any base radius-three recovery set together with the padding generators recovers the full quotient at radius R . Thus

$$R_{\text{full}, R}(s) = s^{\sum_i L_i} R_{\text{base}, 3}(s).$$

The two polynomials remain different by (16)–(17).

Every pair in the construction is realized by an inner code. Put $V = \mathbb{F}_q^J/K$, let $\pi : \mathbb{F}_q^J \rightarrow V$ be the quotient map, and choose an isomorphism $\alpha : \mathbb{F}_q^\ell \rightarrow D/K \leq V$. The generator map

$$G : \mathbb{F}_q^\ell \oplus \mathbb{F}_q^J \longrightarrow V, \quad G(a, x) = \alpha(a) + \pi(x),$$

has helper kernel K and helper preimage of the target span equal to D . Its associated nested pair is therefore $K \subseteq D$. $\square$

The second separation fixes the nested pair itself. Let

$$K = 0, \quad D = \langle u, v \rangle \leq \mathbb{F}_2^3, \quad u = 100, \quad v = 011.$$

The nonzero helper words have weights 1, 2, 3, so $(M_1, M_2) = (1, 3)$.

Proposition 6.3 (The coefficient presentation affects confinement). *Present the target space first by*

$$10 \mapsto u, \quad 01 \mapsto v, \quad 11 \mapsto u + v,$$

and then by

$$11 \mapsto u, \quad 10 \mapsto v, \quad 01 \mapsto u + v.$$

The two inner codes have the same associated nested pair, the same unlabeled set of helper supports, and the same complete relative-weight hierarchy. Their inner-dual weight multisets are respectively $\{2, 3, 5\}$ and $\{3, 3, 4\}$. Hence their values of $M_t(D, K) + d(I^\perp)$, equivalently their eventual confinement thresholds, are

$$(\Gamma_1, \Gamma_2) = (3, 5) \quad \text{and} \quad (\Gamma_1, \Gamma_2) = (4, 6).$$

Proof. Use the graph-code realization

$$I^\perp = \{(-\alpha^{-1}(x), x) : x \in D\},$$

where α is the displayed identification of the two-dimensional target space with D . Adding the target weight to the helper weight gives the two listed multisets directly. Thus $d(I^\perp)$ is two in the first presentation and three in the second. Adding this distance to $(M_1, M_2) = (1, 3)$ gives the two threshold sequences by Theorem 4.2. $\square$

7 The projective-simplex family

The relative-weight hierarchy can be computed exactly for a non-MDS family. Assume $m \geq 2$, put

$$N = \frac{q^m - 1}{q - 1},$$

and let A be an $m \times N$ matrix containing one representative of every point of $\text{PG}(m-1, q)$. Define the inner code by

$$I_m^\perp = \{(a, aA) : a \in \mathbb{F}_q^m\}. \quad (18)$$

Take the first m coordinates as targets and the projective coordinates as helpers. The associated nested pair is isomorphic to

$$0 \subseteq S_m = \{aA : a \in \mathbb{F}_q^m\},$$

where S_m is the q -ary simplex code. The generalized-weight formula for S_m is classical; in fact every t -dimensional simplex subcode has the same support size [16]. We include the short projective count to connect that hierarchy to the confinement and reliability statements below.

Theorem 7.1 (Projective-simplex thresholds). *For $1 \leq t \leq m$,*

$$M_t(S_m, 0) = \frac{q^m - q^{m-t}}{q - 1}, \quad (19)$$

and

$$d(I_m^\perp) = q^{m-1} + 1. \quad (20)$$

Hence

$$\Gamma_t := M_t(S_m, 0) + d(I_m^\perp) = \frac{q^m - q^{m-t}}{q - 1} + q^{m-1} + 1. \quad (21)$$

In outer families with dual distance tending to infinity, Γ_t is the eventual first nonconfined helper cost at recovered dimension t . The family is non-MDS for $m \geq 2$; for $m \geq 3$ the successive increments of Γ_t are unequal.

Proof. The target restriction of a word $(a, aA) \in I_m^\perp$ is a , and no nonzero dual word has zero target restriction. Hence normalized recovery systems correspond, with support preserved, to subcodes of S_m ; this proves the asserted identification of the associated pair.

Let $U \leq \mathbb{F}_q^m$ have dimension t . The common zero coordinates of the subcode UA are the projective points in $U^\perp$, a subspace of projective dimension $m - t - 1$. Thus

$$|\text{supp}(UA)| = N - \frac{q^{m-t} - 1}{q - 1} = \frac{q^m - q^{m-t}}{q - 1},$$

independently of U . This proves (19). Every nonzero simplex word aA has weight q^{m-1} , while a nonzero target vector a has weight at least one. Equality is attained by a coordinate vector, which proves (20). The confinement formula is Theorem 4.2. The non-MDS and increment assertions follow from the displayed parameters. $\square$

This family also gives an exact stochastic law at every recovered dimension. Let each helper survive independently with probability s , and let F be the set of failed projective points. The available recovery equations are indexed by

$$\{a \in \mathbb{F}_q^m : aA|_F = 0\} = (\text{span } F)^\perp,$$

which has dimension $m - \text{rank}(F)$.

Theorem 7.2 (Projective rank reliability). *The probability of retaining at least t independent target recovery equations is*

$$R_t(s) = \sum_{\substack{F \subseteq \text{PG}(m-1, q) \\ \text{rank}(F) \leq m-t}} (1-s)^{|F|} s^{N-|F|}. \quad (22)$$

Writing $N_v = (q^v - 1)/(q - 1)$, this equals

$$R_t(s) = \sum_{u=0}^{m-t} \begin{bmatrix} m \\ u \end{bmatrix}_q \sum_{v=0}^u \begin{bmatrix} u \\ v \end{bmatrix}_q (-1)^{u-v} q^{\binom{u-v}{2}} s^{N-N_v}. \quad (23)$$

Its endpoint terms are

$$R_t(s) = \begin{bmatrix} m \\ t \end{bmatrix}_q s^{M_t} + O(s^{M_t+1}) \quad (s \rightarrow 0), \quad (24)$$

$$1 - R_t(s) = B_{m, m-t+1} (1-s)^{m-t+1} + O((1-s)^{m-t+2}) \quad (s \rightarrow 1), \quad (25)$$

where

$$B_{m,r} = \begin{bmatrix} m \\ r \end{bmatrix}_q \frac{|\text{GL}(r, q)|}{r!(q-1)^r}$$

is the number of unordered projective r -frames.

Proof. Recovery of t independent target combinations is possible exactly when $m - \text{rank}(F) \geq t$, which gives (22). For a vector subspace $U \leq \mathbb{F}_q^m$, let $\mathbb{P}(U)$ denote its projective points. Möbius inversion in the subspace lattice gives the total probability of failed sets spanning a fixed u -space U as

$$\sum_{V \leq U} \mu(V, U) s^{N-N_{\dim V}}, \quad \mu(V, U) = (-1)^{u-v} q^{\binom{u-v}{2}}, \quad v = \dim V.$$

Here the total weight of all failed sets contained in $\mathbb{P}(V)$ is

$$\sum_{F \subseteq \mathbb{P}(V)} (1-s)^{|F|} s^{N-|F|} = s^{N-N_{\dim V}},$$

which accounts for the power of s before inversion. There are $\begin{bmatrix} m \\ u \end{bmatrix}_q$ choices for U and $\begin{bmatrix} u \\ v \end{bmatrix}_q$ subspaces $V \leq U$ of dimension v . Summing over $u \leq m-t$ proves (23).

For $s \rightarrow 0$, a smallest surviving set that permits rank- t recovery is the complement of the points in an $(m-t)$ -dimensional vector subspace. It has size M_t , and there are $\begin{bmatrix} m \\ m-t \end{bmatrix}_q = \begin{bmatrix} m \\ t \end{bmatrix}_q$ such subspaces. For $s \rightarrow 1$, failure first occurs when the failed points span dimension $m-t+1$. A smallest such set is a projective frame of that size. Counting its span and then its unordered projective bases gives $B_{m,m-t+1}$. $\square$

At the same helper length, the simplex presentation is characterized by its first recovery cost. This is the length- N , multiplicity-one specialization of Bonisoli's classification of linear equidistant codes [2]; the proof below records the precise zero-column and projective-multiplicity conventions used here.

Proposition 7.3 (Equality in the first projective bound). *Let B be any full-row-rank $m \times N$ matrix over $\mathbb{F}_q$, with $N = (q^m - 1)/(q - 1)$. Then*

$$\min_{0 \neq a \in \mathbb{F}_q^m} \text{wt}(aB) \leq q^{m-1}.$$

Equality holds if and only if, up to column scaling and permutation, the columns of B represent every point of $\text{PG}(m-1, q)$ exactly once.

Proof. For each nonzero column, the fraction of nonzero $a \in \mathbb{F}_q^m$ for which its entry in aB is nonzero is

$$\frac{q^{m-1}(q-1)}{q^m-1}.$$

Averaging $\text{wt}(aB)$ over $a \neq 0$ gives at most q^{m-1} , strictly less if B has a zero column. If the minimum attains this average, every nonzero linear combination has weight q^{m-1} and every projective hyperplane contains total column multiplicity $(q^{m-1} - 1)/(q - 1)$.

Let c be the vector of projective column multiplicities. The point-hyperplane incidence matrix is square, and its real Gram matrix has the form $(r - \lambda)I + \lambda J$ with $r > \lambda$. It is nonsingular, so the constant hyperplane-sum equations have the unique solution $c = \mathbf{1}$. Conversely, the simplex matrix has constant nonzero weight q^{m-1} . $\square$

8 Verification boundary

The paper has a paper-owned Lean 4 companion built against a pinned Mathlib revision. It formalizes the construction of the associated nested code pair from abstract target and helper generator maps. In the notation of (1), the checked statements are

$$K_P \leq D_P, \quad G_J(D_P) = W_P, \quad G_J|_{D_P} : D_P \twoheadrightarrow W_P, \quad \ker(G_J|_{D_P}) = K_P.$$

The corresponding declarations are

Paper statement	Lean declaration
$K_P \leq D_P$	<code>helper_ker_le_helperCodeForTargetSpace</code>
$G_J(D_P) = W_P$	<code>map_helperCodeForTargetSpace_eq_recoverableTargetMessageSpace</code>
surjectivity	<code>recoverableTargetMap_surjective</code>
restricted kernel	<code>mem_ker_recoverableTargetMap_iff</code>

The companion uses Lean 4.32.0-rc1 and the Mathlib commit

571b8a8e54219b4d393f75f4b8653fac08197fcc.

The axiom audit of all four reviewer-facing declarations reports only

`{Classical.choice, Quot.sound, propext}`.

No coding-theory result is introduced as a Lean axiom.

The formal coverage stops at the exact sequence. In particular, Theorems 3.2, 4.1, 4.2, and 4.3, as well as the strictness construction in Proposition 4.4 and the later consequences and examples, are human-proved and are not established by the paper-local Lean companion. The claim manifest records them as absent rather than treating the exact-sequence formalization as evidence for them.

The reliability polynomials in Proposition 6.1 follow from the displayed union-size table by inclusion–exclusion. No exhaustive computation is a premise of that proposition or of any theorem in the main proof chain. Machine-readable examples and replay checks may therefore be used to audit arithmetic without changing the logical dependence of the paper.

9 Conclusion

For a fixed target set, the associated nested code pair separates two parts of linear recovery. Its relative generalized Hamming weights give the exact minimum helper unions for recovered subspaces of each dimension. The full normalized recovery equations retain the coefficient and overlap data that those minima discard.

The dual nested pair describes the same recovery problem from the failure side: its relative weights are the minimum numbers of helper failures that leave prescribed dimensions of target ambiguity. Thus helper cost, erasure ambiguity, and rank reliability are different summaries of the recovered dimension as the surviving helper set varies.

After the outer-distance gate, concatenation turns the relative-weight hierarchy into an additive hierarchy of exact confinement thresholds:

$$M_t(D_P, K_P) + d(I^\perp).$$

In sufficiently long concatenations whose outer dual distance grows, every bounded equation below the corresponding threshold remains in its inner block and zero-extension copies its coefficients and exact support. This yields positive-density realizations and eventually transfers any consequence that depends only on the copied minimal recovery sets, including bounded repair reliability and service-rate regions.

At one target coordinate in a block with nonzero outer projection, retaining the nonzero functional-dual fibers removes that gate. The target-constrained inner coset weights give the exact finite nonconfinement cost, and the Singer-cycle construction shows that this weighted criterion can certify transfer when ordinary outer support distance cannot. Thus the paper gives arbitrary-rank exactness after the block-distance gate and a finer ungated finite-length theorem in rank one.

The two separations locate the remaining information. Equal relative weights do not determine the stochastic law because they do not retain the intersection pattern of all recovery sets. Even the associated nested pair does not determine the additive threshold without the target coefficient presentation, which changes $d(I^\perp)$. The projective-simplex family exhibits both levels explicitly: its relative weights give the minimum recovery costs, while the projective rank distribution gives the complete dimension-by-dimension reliability law.

Two extensions require different objects. Disjoint simultaneous service is an integral packing problem, rather than a minimum-union or fractional service-rate problem. Bandwidth-aware repair for array codes allows each helper to transmit a subspace of functionals, so its cost is not Hamming support. Neither extension follows by changing terminology in the present theorems; each requires a new confinement invariant adapted to its cost model.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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